

Computational Fluid Dynamics

Prof. Parolini and Prof. Valdettaro

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We want to simulate the flow through the divergent duct sketched in Figure 1. The problem is governed by the time-dependent incompressible Navier-Stokes equations in total stress (rotational) formulation:

$$\begin{cases} \frac{\partial \mathbf{u}}{\partial t} - \nu \Delta \mathbf{u} + \boldsymbol{\omega} \times \mathbf{u} + \nabla p_T = 0, & \text{in } \Omega, \quad t > 0 \\ \operatorname{div} \mathbf{u} = 0, & \text{in } \Omega, \quad t > 0, \\ \mathbf{u} = 0, & \text{in } \Omega, \quad t = 0, \end{cases} \quad (1)$$

where $\mathbf{u}$ is the fluid velocity, $p_T = p + |\mathbf{u}|^2/2$ is the total pressure, $\boldsymbol{\omega} = \nabla \times \mathbf{u}$ is the vorticity and $\nu = 0.1$ is the kinematic viscosity. The problem is simulated only on the lower half of the domain exploiting the symmetry and the boundary conditions are reported in Figure 1.

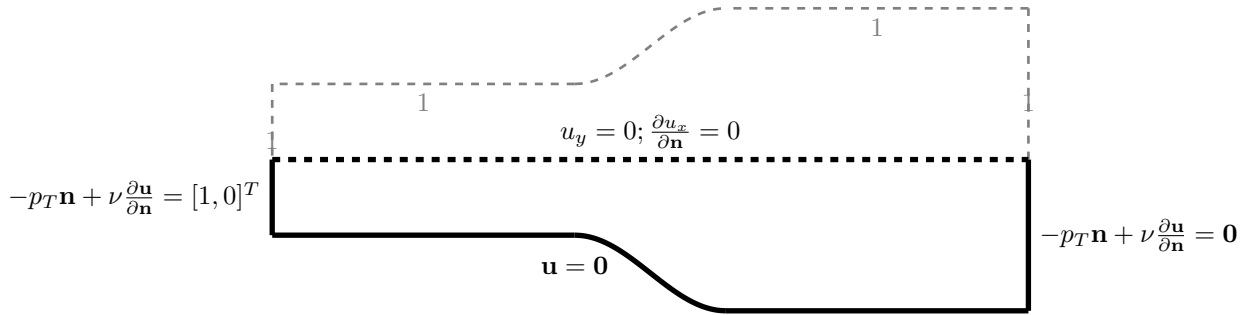


Figure 1: Computational domain and boundary conditions

- Write the spatial semi-discretization of Problem (1) based on the finite element method.
- Introduce a time-discretization based on the backward Euler scheme with a semi-implicit treatment of the convective term ($\boldsymbol{\omega}^n \times \mathbf{u}^{n+1}$).
- Based on the template `exam.edp`, write a `FreeFem++` script for the solution of Problem (1) using $\mathbb{P}_2 - \mathbb{P}_1$ finite elements for velocity and pressure, respectively, starting from the fluid at rest.
- Compute the solution on the supplied grid using a time step $\Delta t = 0.2$ until the final time $T = 10$. Evaluate the flowrate through the duct and the force acting on the lower wall at the final time.
- Is the stability of problem (1) guaranteed for arbitrary boundary conditions?